

Dani R. Lipman

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RESEARCH FOCUS

I am interested in exploring dynamics and 3-D modeling of galaxy centers using both observations and MHD simulations. I am particularly interested in understanding how material may fall off nuclear orbits and inflow closer to circum-nuclear discs near central supermassive black holes. My main question is: How do dynamics and physical parameters work together to control star formation and galaxy evolution?

EDUCATION

- **University of Connecticut** August 2020 - Present
Physics Ph.D Candidate Storrs, CT
 - Expected graduation May 2025
 - MSc in Physics conferred May 2023
- **University of Iowa** 2015 - 2019
Undergraduate Education Iowa City, IA
 - B.S. in Physics and Astronomy (Double Major), Mathematics Minor
 - Received University Honors; Honors in Astronomy

COLLABORATIONS

- **ALMA CMZ Exploration Survey (ACES)** 2022 - Present
Google site: <https://sites.google.com/view/aces-cmz/home>
 - Band 3 ALMA large program completed in Cycle 8 (2021.1.00172.L). observed the entire CMZ at 3mm at spatial resolutions (FWHM) of $\sim 2''$. Goal to measure global mass flow.
 - Active participant in theory working group. Goal to use zoom-in MHD simulation for direct comparison with ACES
 - Led and participated in multiple intensive paper sprints to perform analysis and produce a full paper draft with a small group within 2-weeks.
 - Leading the development of solar mass resolution Zoom-in Simulations of Milky-Way like galaxies using the moving mesh code AREPO. Simulations combine updated external gravitational potential from Hunter et al. (2024), magnetic fields, star particles [representing clusters], accreting sink particles [representing prestellar cores], and supernova feedback.
- **3-D Central Molecular Zone Collaboration** 2020 - Present
3-D CMZ Paper Series: https://centralmolecularzone.github.io/3D_CMZ/
 - Work included as part of associated paper series, which explores global properties and hierarchical structure of the CMZ.
 - Created a Bayesian framework to synthesize existing datasets for Near/Far position distinctions in the CMZ.
 - Developed 4-D (ℓ, b, v and near/far) fitting method of an x2 elliptical orbit for the Milky Way's CMZ.

RESEARCH POSITIONS

2024 - Present	NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellow <i>Ph.D. Advisor: Associate Prof. Cara Battersby</i>
2020 - 2024	Graduate Research Assistant (UConn) <i>Graduate advisor: Associate Prof. Cara Battersby</i>
2019 - 2020	China Fulbright Study/Research Fellow (Nanjing, China) <i>Investigating Perception & Design of Science Outreach in Chinese Universities</i>
2016 - 2019	Undergraduate Research Assistant (UIowa; Prof. Jasper Halekas) <i>Characterizing the Structure of the Martian Bow Shock</i>
Summer 2017	Technion Astrophysics Research Group (Technion Israel Institute of Technology; Prof. Hagai Perets) <i>Two introductory summer projects: (i) Pollution of White Dwarf Binaries by Exocomets, (ii) Radio Occultation of Exoplanets</i>
2016 - 2020	Undergraduate Research Assistant (UIowa; Prof. Robert Mutel) <i>Search for thermal gyro-synchrotron emission in pre-main sequence and T-Tauri stars</i>

SELECTED PUBLICATIONS

- [1] **Lipman, D.**, Battersby, C., Walker, D. L., et al. 2025, The Astrophysical Journal, 984, 159, doi: [10.3847/1538-4357/adb5ee](https://doi.org/10.3847/1538-4357/adb5ee)
- [2] Battersby, C., Walker, D. L., Barnes, A., et al. 2025b, The Astrophysical Journal, 984, 156, doi: [10.3847/1538-4357/adb5f0](https://doi.org/10.3847/1538-4357/adb5f0)
- [3] Walker, D. L., Battersby, C., **Lipman, D.**, et al. 2025, The Astrophysical Journal, 984, 158, doi: [10.3847/1538-4357/adb5ef](https://doi.org/10.3847/1538-4357/adb5ef)
- [4] Tress, R. G., Sormani, M. C., Girichidis, P., et al. 2024, AA, 691, A303, doi: [10.1051/0004-6361/202450035](https://doi.org/10.1051/0004-6361/202450035)
- [5] Brunker, S. W., Battersby, C., Albolani, D., et al. 2025, , 982, L20, doi: [10.3847/2041-8213/adba5e](https://doi.org/10.3847/2041-8213/adba5e)
- [6] Albolani, D., Battersby, C., Brunker, S. W., et al. 2025, , 169, 213, doi: [10.3847/1538-3881/adba3f](https://doi.org/10.3847/1538-3881/adba3f)
- [7] Sofue, Y., Oka, T., Longmore, S. N., et al. 2025, , Publications of the Astronomical Society of Japan, 2025, doi: [10.1093/pasj/psaf072](https://doi.org/10.1093/pasj/psaf072)

SELECTED CONFERENCES AND WORKSHOPS

Sept 2025	Stellar Origins Conference (Vienna, Austria) – Poster
July 2025	New England Star and Planet Formation Meeting (Yale University) – Talk
Sept 2023	Arepo Development Workshop for ISM simulations – Invited attendee for collaborative development workshops for new simulation modules
June 2023	New England Star and Planet Formation Meeting (Harvard CfA) – Poster
Apr 2023	Protostars and Planets VII (Kyoto, Japan) – Poster
Sept 2022	Galactic Centre Puzzles (Heidelberg, Germany) – Invited participant for discussions
June 2022	New England Star and Planet Formation Meeting (Wesleyan University) – Talk

INVITED TALKS, POSTERS, & PANELS

Jan 2025	Conference for Undergraduate Women in Physics (UConn) – Graduate Student Panelist
Apr 2024	EcoTarium Museum of Science and Nature (Boston, MA) – Invited Public Talk
Feb 2024	Kansas University Astronomy Seminar – Invited Talk
Oct 2024	UConn Women in Mathematics and Applications Seminar – Invited Talk
July 2023	MIT Haystacks Astronomy Community Day – Poster
Nov 2022	Connecticut Space Grant Consortium Expo – Poster
Nov 2018	Iowa Space Grant Consortium Expo – Poster
Jan 2018	Conference for Undergraduate Women in Physics (Ames, IA) – Poster
Nov 2017	Iowa Space Grant Consortium Expo – Talk
Sept 2016	University of Iowa ICRU grant poster session – Poster

SKILLS

- **Programming Languages:** Python (advanced), C (proficient), IDL (proficient), basic html
- **Visualization Tools:** Proficient use of DS9, CARTA, glueviz, and 3-D python visualization libraries
- **Web Technologies & Version Control:** Proficient use of GitHub, BitBucket
- **Database Systems:** Globus, Harvard Dataverse, IRSA
- **Specialized Area:** Development of MHD simulations using AREPO code suite, extensive use of hdf5 and binary filetypes
- **High Performance Computing:** Proficient use of high-performance computing and slurm. Introductory use of docker and singularity container platforms
- **Personal Skills:** Intermediate speaking/reading level Mandarin Chinese

HONORS AND AWARDS

2024	NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellow Title: <i>Central Station: Mass Transport into the Galactic Center with AREPO</i>
2022	Connecticut Space Grant Consortium Graduate Research Fellowship Title: <i>3D CMZ: Distinguishing Near vs. Far Distances in the Galactic Center Using Spitzer and Herschel</i>
2019	China Fulbright Study/Research Award Title: <i>Investigating Perception and Design of Science Outreach in Chinese Universities</i>
2019	Fulbright Critical Language Enhancement Award <i>3-month intensive language immersion project prior to starting the Study/Research award. Completed with Advanced speaking level.</i>
2019	UIowa Physics Department Distinguished Service Award
2017	University of Iowa Latham Science Engagement Initiative Fellow <i>Yearlong course to develop science outreach programs. Developed group and individual outreach projects aimed at high school and general audiences, respectively.</i>
2017; 2018	Iowa Space Grant Consortium Undergraduate Research Fellowship Title: <i>Characterizing the Structure of the Martian Bow Shock</i>
2018	UIowa Physics Department James A. Van Allen Award
2017	UIowa Physics Department Outstanding Service Award

LEADERSHIP AND OUTREACH EXPERIENCE

- **UConn Astronomy Seminar Committee**

2022 - Present

- **Collaboration journal club** 2022 - Present
 - Lead organizer of virtual monthly journal club between three US-based research groups. Focused on graduate student-lead presentation of timely research papers.
- **Research Mentorship of Undergraduate Students** 2021 – Present
 - Jakub Poznanski (Grad Spring '26)
 - John Trujillo (Grad Spring '25)
 - Sangeeta Kuchibhotla (Grad Spring '24)
- **Co-Organizer of Astronomy on Tap Storrs** [🌐] 2022 – 2025
- **Co-lead of UConn STARS: Science Technology and Astronomy Recruits** 2022 - 2024
 PI: Prof. Cara Battersby (UConn) [🌐]
 - Program for undergraduate students in physics, aimed to recruit and retain students from historically excluded groups in physics.
 - Organized and led undergraduate in the creation and teaching of lesson plans for under-resourced k-12 schools
- **Hartford Public Library Outreach Events** 2022 - 2024
 - Organized and lead 4 after school programs for k-6 aged children

TEACHING EXPERIENCE

- **Graduate Teaching Assistant** Fall 2020 – Fall 2022
 University of Connecticut Department of Physics
 - Assisted and led active-learning classroom lectures, discussion, and labs for Physics II for engineering students
 - Taught traditional 3-hour labs for algebra-based Physics I
- **Undergraduate Teaching Assistant** Fall 2017 – Spring 2019
 University of Iowa Department of Physics & Astronomy
 - Assisted in active-learning classroom lectures, and led discussion sections for pre-med focused Physics II, as well as Physics I and Physics II for engineering students.
- **Honors Writing Fellow** Fall 2016 – Fall 2017
 University of Iowa Writing Center
 - Fellowship position where awardees are assigned to a professor for a semester to peer review midterm and final papers. Met with students individually to discuss drafts.
- **Writing Center Tutor** Fall 2017 – Spring 2019
 University of Iowa Writing Center
 - Tutored scheduled appointments with undergraduate and graduate students from any major throughout the university.